Subfield interpolation lowers the T-count of GF(2^n) multipliers by up to half

Claude Opus 5.5 Extra Effort[1]

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We report ancilla-free Clifford+T circuits for multiplication in the binary fields GF(2^n) that use fewer T gates than the best published ones for $n = 8$, 12, 14, 15, 16, and 32: 117, 195, 309, 261, 267, and 585 T gates, against previous records of 123, 243, 319, 363, 383, and 1177. The records came from reinforcement learning, from TODD-type reduction, and from LLM-guided evolutionary search. Our circuits come from a simple composition rule. If GF(2^{md}) is a degree-d extension of $K = \mathrm{GF}(2^m)$ and $2^m \geq 2d-2$, then multiplication by interpolation over K needs $2d-1$ products in K. Each product is a linear relabelling of a multiplier for K, so the T-count obeys $\mathcal{T}(md) \leq (2d-1)\,\mathcal{T}(m)$. Starting from published small-field decompositions, this gives all six records without search, and one subsequent TODD-type reduction step lowers $n = 16$ from 273 to 267. Combined with published small-field values, the same rule implies further bounds, such as $\mathcal{T}(10) \leq 147$ and $\mathcal{T}(12) \leq 161$, which we state separately because we did not verify those bases. We also prove by exhaustive enumeration that 17 T gates are optimal for GF(4) multiplication. Every circuit was certified by exact arithmetic in two independent implementations. The results were found by a multi-agent language-model system working against a frozen, test-validated verifier, and we describe its safeguards.

I. INTRODUCTION

In fault-tolerant quantum computing with magic-state distillation, the T gate is the expensive gate, and its count is a standard cost metric [1, 2]. Minimizing the T-count of a circuit is hard in general. For circuits whose non-Clifford part is a phase polynomial it is equivalent to decoding a punctured Reed–Muller code, or to finding a minimal symmetric (Waring) decomposition of a third-order tensor over $\mathbb{F}_2$ [3, 4]. Heuristics for this problem include TODD [4] and its descendants FastTODD [5], reinforcement learning (AlphaTensor-Quantum) [6], tensor-decomposition methods [7], and LLM-guided evolutionary search (VarTODD+GigaEvo) [8]. Together they define the state of the art on standard benchmark circuits [9].

Multiplication in GF(2^n) is one of these benchmarks and a building block of quantum attacks on binary elliptic curves [10–12]. Its best published T-counts (Table I) come from three sources. Vandaele's FastTODD applied to Karatsuba-type circuits gives the best value for $n = 3$ [5, 13]. AlphaTensor-Quantum gives the best for $n = 4$, 6 and 8 [6], with Fisher *et al.* matching these. VarTODD+GigaEvo gives the best for $n = 5$, 7, 9–16 and 32 [8]; the value 101 for $n = 7$ was also reached in Ref. [7].

Here we observe that these records ignore the structure of the field. For composite $n = md$, GF(2^n) is an extension of its subfield GF(2^m). Multiplication in the extension reduces to $2d-1$ subfield multiplications when the subfield has enough points (Toom–Cook interpolation, or Karatsuba's formula for $d = 2$ [14–16]). We show that the T-count composes accordingly. Every subfield product is a relabelling of a GF(2^m) multiplier, so a good small-field decomposition becomes a good large-field one. The resulting circuits beat all published values for six field sizes, by up to a factor of 2 (Table I). Section II fixes notation. Section III proves the composition rule. Section IV presents the records, the $n = 2$ optimality proof, and the bounds implied by published values. Section V describes the verification, and Section VI how the results were found.

TABLE I. T-counts of ancilla-free GF(2^n) multipliers. "Previous" is the best published value: Ref. [8] (Table 1) for $n \geq 12$, and Refs. [6, 8] for $n = 8$. "This work" lists certified circuits: R products in the subfield $K = \mathrm{GF}(2^m)$ of degree d, with base decompositions from Ref. [6] ($m = 3, 4$) or our own ($m = 7$; see text). *Followed by a FastTODD-type reduction from 273 to 267.

n	Previous	This work	$R \times \mathcal{T}(m)$	K, d
8	123	**117**	3×39	GF(16), 2
12	243	**195**	5×39	GF(16), 3
14	319	**309**	3×103	GF(2^7), 2
15	363	**261**	9×29	GF(8), 5
16	383	**267**	$7 \times 39{=}273^*$	GF(16), 4
32	1177	**585**	15×39	GF(16), 8

II. PRELIMINARIES

Let $P \in \mathbb{F}_2[x]$ be irreducible of degree n. Elements $a, b, c \in \mathrm{GF}(2^n) = \mathbb{F}_2[x]/(P)$ are coordinate vectors in $\mathbb{F}_2^n$. The in-place multiplier $|a, b, c\rangle \mapsto |a, b, c \oplus ab\rangle$ equals $H_c\, D\, H_c$, where H_c applies Hadamard gates to the c register and D is the diagonal gate

$$D = \sum_x (-1)^{f_n(x)} |x\rangle\langle x|, \qquad f_n(a, b, c) = c \cdot (ab), \quad (1)$$

where $x = (a, b, c)$ and $\cdot$ is the dot product over $\mathbb{F}_2$. The function f_n is a cubic form in the $N = 3n$ variables of x,

whose monomials are $a_i b_j c_k$ for each x^k that appears in $x^{i+j} \bmod P$.

A T gate applied to the parity $u \cdot x$ (a phase gadget, conjugated by CNOT gates) contributes the phase $\omega^{\ell_u(x)}$, with $\omega = e^{i\pi/4}$ and $\ell_u(x) \in \{0,1\}$ the parity as an integer. A set $\{u_1, \dots, u_m\} \subset \mathbb{F}_2^N \setminus \{0\}$ *implements* D if D equals $\prod_r \omega^{\ell_{u_r}}$ times a diagonal Clifford gate. The T-count $\mathcal{T}(n)$ is the least such m. It does not depend on P or on the basis, because those changes are CNOT circuits.

Lemma 1 (Signature-tensor criterion [3, 4])**.** *$\{u_r\}$ implements D if and only if every variable and every pair of variables lies in the support of an even number of the u_r, and every triple lies in an odd number of them exactly when it is a monomial of f_n. Equivalently, $\sum_r u_r^{\otimes 3} \equiv S_n \pmod 2$, where S_n is the zero-diagonal symmetric tensor of f_n.*

Proof. Over $\mathbb{Z}_8$, $\ell_u(x) = \sum_{i\in u} x_i - 2\sum_{i<j\in u} x_i x_j + 4\sum_{i<j<k\in u} x_i x_j x_k$, and terms of degree ≥ 4 vanish because their coefficients are multiples of 8. Hence $4f_n - \sum_r \ell_{u_r}$ has linear coefficients $-\#\{r \ni i\}$, quadratic coefficients $2\#\{r \ni i,j\}$ and cubic coefficients $4(f_{ijk} - \#\{r \ni i,j,k\})$. A diagonal phase $\omega^{g(x)}$ is Clifford exactly when $g = 2h$ with h of degree ≤ 2 over $\mathbb{Z}_4$ and even quadratic coefficients (products of S and CZ gates). That is, the singles and pairs must be even and the triples must match $f_n \bmod 2$. □

III. COMPOSITION RULE

Lemma 2 (Pull-back)**.** *Let $\{u_r\}$ implement the cubic form $g(y)$ on $\mathbb{F}_2^M$, and let $L : \mathbb{F}_2^N \to \mathbb{F}_2^M$ be linear. Then $\{L^{\mathsf{T}} u_r\}$, after deleting zero vectors and cancelling equal pairs, implements $g(Lx)$. Its size is at most the size of $\{u_r\}$.*

Proof. $\ell_u(Lx) = \ell_{L^{\mathsf{T}}u}(x)$, and Clifford gates on $y = Lx$ pull back to Clifford gates in x. Zero vectors and pairs u, u contribute Clifford phases ($\omega^{2\ell_u} = i^{\ell_u}$). □

Theorem 1. *Let $n = md$, $K = \mathrm{GF}(2^m) \subset \mathrm{GF}(2^n)$, and suppose multiplication in $\mathrm{GF}(2^n)$, viewed as a d-dimensional K-algebra, has a K-bilinear formula $ab = \sum_{r=1}^{R} \alpha_r(a)\,\beta_r(b)\,X_r$ with K-linear forms α_r, β_r and constants $X_r \in \mathrm{GF}(2^n)$. Then $\mathcal{T}(n) \leq R\,\mathcal{T}(m)$.*

Proof. Write $f_n(a,b,c) = \sum_r c \cdot (\alpha_r(a)\beta_r(b)X_r)$. For fixed r and c, the map $p \mapsto c\cdot(pX_r)$ is $\mathbb{F}_2$-linear in $p \in K$, so it equals $\gamma_r(c)\cdot p$ for some $\mathbb{F}_2$-linear γ_r. Hence $f_n = \sum_r f_m(\alpha_r(a), \beta_r(b), \gamma_r(c))$, where f_m is the cubic form of K-multiplication with a fixed nondegenerate pairing. Lemma 2 applied to each term, with $L_r = (\alpha_r, \beta_r, \gamma_r)$, gives a decomposition of size $\leq R\,\mathcal{T}(m)$. The union of the pulled-back sets implements the sum, because phases multiply and Clifford gates compose. □

Corollary 1 (Interpolation)**.** *If $2^m \geq 2d-2$, then $\mathcal{T}(md) \leq (2d-1)\,\mathcal{T}(m)$. For $d = 2$ this holds for every m (Karatsuba, $R = 3$).*

Proof. Write $a = \sum_{j<d} A_j y^j$ and $b = \sum_{j<d} B_j y^j$ in a K-basis $1, y, \dots, y^{d-1}$ of $\mathrm{GF}(2^n) = K[y]/(g)$, with g irreducible of degree d over K. The product $A(y)B(y)$ has degree $2d-2$ and is determined by its values $A(t)B(t)$ at $2d-2$ distinct points $t \in K$ together with its leading coefficient $A_{d-1}B_{d-1}$ (the point at infinity) [15, 16]. Lagrange interpolation followed by reduction modulo g is K-linear, which gives a formula with $R = 2d-1$. □

It is classical that $2d-1$ is the least possible number of K-products, and that it is achievable precisely when $2^m \geq 2d-2$ [16, 17]. The asymptotic linearity of bilinear complexity over $\mathbb{F}_2$ [18, 19], together with $\mathcal{T} \leq 7\times$(number of CCZ gates), already implies $\mathcal{T}(n) = O(n)$. The point here is the constant: Theorem 1 multiplies the subfield's *T-count*, not $7\times$ its bilinear rank. For $n = 16$, $7 \times \mathcal{T}(4) = 273$, whereas $7 \times (7 \cdot 9) = 441$.

IV. RESULTS

Records. Table I lists the certified circuits. Five use Corollary 1 with base decompositions of GF(8) (29 T gates) and GF(16) (39 T gates) published by AlphaTensor-Quantum [6]. For $n = 8$ we used its Karatsuba form, $d = 2$ over GF(16). The sixth, $n = 14$, uses a 103-gate decomposition of $\mathrm{GF}(2^7)$ that we obtained by FastTODD-type reduction of the published 104-gate one. The current $\mathrm{GF}(2^7)$ record is 101 [7, 8], so with that decomposition one would get 303. For $n = 16$, a FastTODD-type pass on the 273-gate interpolation circuit removes six more T gates, giving 267. The interpolation points are the first $2d-2$ elements of K in coordinate order plus infinity, and the moduli are listed in Appendix A. All constructions are deterministic and take well under a second.

Optimality for $n = 2$. The decompositions of a fixed signature tensor form a coset of the code $C = \{w \in \mathbb{F}_2^{2^N-1} : \sum_y w_y y^{\otimes 3} = 0\}$, which is the punctured Reed–Muller code $\mathrm{RM}(N-4, N)^*$ [3]. For GF(4) ($N = 6$), C has dimension 22. Enumerating all 2^{22} coset elements gives minimum weight 17, attained by 1008 decompositions, with the next weights 18 (2576) and 21 (24 816). Hence $\mathcal{T}(2) = 17$, which the published circuits already attain [5, 6]. A second implementation, written independently of the first, reproduced these numbers.

Implied bounds. Theorem 1 also applies to published small-field values. We did not have those decompositions and did not verify them, so we list the resulting bounds separately (Table II). Notably, Ref. [8] reports $\mathcal{T}(5) = 49$ and $\mathcal{T}(10) = 157$; by Corollary 1 with $d = 2$, the former implies $\mathcal{T}(10) \leq 147$. Bounds that use only our certified bases are fully rigorous. An example is $\mathcal{T}(64) \leq 15 \times$

TABLE II. Bounds implied by Corollary 1. "Published base" bounds rely on decompositions reported in Refs. [5, 7, 8] that we have not verified: 23 for $n = 3$ [5], 49 for $n = 5$ [8], and 101 for $n = 7$ [7, 8]. The last two rows use only bases we have verified.

n	Previous	Bound	Route	Base
6	77	69	$3 \times \mathcal{T}(3)$, $d = 2$	published
9	139	115	$5 \times \mathcal{T}(3)$, $d = 3$	published
10	157	147	$3 \times \mathcal{T}(5)$, $d = 2$	published
12	243	161	$7 \times \mathcal{T}(3)$, $d = 4$	published
14	319	303	$3 \times \mathcal{T}(7)$, $d = 2$	published
15	363	207	$9 \times \mathcal{T}(3)$, $d = 5$	published
20	–	351	$9 \times \mathcal{T}(4)$, $d = 5$	ours
64	3729	1755	$15 \times \mathcal{T}(8)$, $d = 8$	ours

$117 = 1755$ ($d = 8$ over GF(256)), against the published 3729 [8].

Where it does not help. Prime n (5, 7, 11, 13) has no proper subfield. For $n \leq 7$ and for $n = 9$ and 10 our own searches matched but did not beat the published values. FastTODD-type reduction of the published decompositions reached $n = 7$: 103, $n = 9$: 161, and $n = 10$: 191. Simulated annealing over the coset of decompositions reached $n = 4$: 39 from a schoolbook start in about a second, never fewer. The rank-6 bilinear formulas for GF(8) that we reduced (25 of the 147) all ended at 29, not 23. Better small bases would propagate through Theorem 1 to every composite n.

V. VERIFICATION

Frozen verifier. A candidate lists n, an irreducible modulus and the vectors u_r. The verifier checks irreducibility (Rabin's test) and then the criterion of Lemma 1 exactly, using the parities of all singles, pairs and triples. A candidate wins only if it has fewer vectors than the record. Before the search started, the verifier's sources, the record table and the compiled binary were pinned by hash; they were re-pinned after the record update described in Sec. VI. The verifier accepts all nine published AlphaTensor-Quantum decompositions for $n = 2$–10. It accepts schoolbook CCZ constructions, rejects malformed input, and exercises every verdict path. In random property tests, 80 out of 80 single-bit corruptions were rejected, and 80 out of 80 "zero-sum moves" were accepted. A zero-sum move toggles the 15 nonzero vectors of a random 4-dimensional subspace, which leaves $\sum u^{\otimes 3}$ unchanged.

Independent circuit check ($n \leq 8$). A separately written C program evaluates $4f_n(x) - \sum_r \ell_{u_r}(x) \bmod 8$ for all 2^{3n} inputs. It computes the sum through a Walsh–Hadamard transform and checks that the remainder is a diagonal Clifford phase, via a Möbius transform over $\mathbb{Z}_4$. This certifies the circuit directly rather than through the tensor criterion. The two checks agree on every test, and disagreement would have raised an alarm. The $n = 8$ record passes this check.

Independent tensor check (all n). A third implementation, using numpy and sharing no code with the verifier, builds $\sum_r u_r^{\otimes 3}$ and the target tensor from scratch. For each record it finds zero mismatches among all N^3 entries, for instance 884 736 entries at $n = 32$. The multi-agent auditor (Sec. VI) also re-checked, with its own implementations, every record except $n = 16$: 267 and $n = 32$: 585, which arrived after its audit. One of these is a randomized circuit-level test that fits the $\mathbb{Z}_4$ Clifford remainder and checks it on 2×10^5 random inputs and all weight-3 inputs, for n up to 16. Negative controls with one flipped bit fail every check.

Novelty. Our record table (Table I) combines Refs. [5, 6, 8]. Ref. [7] reports T-counts only for $n \leq 10$, and its values do not improve on these. We found no published T-count at or below ours for the six field sizes. Tower and interpolation constructions are classical in algebraic complexity [14–18]. We do not claim them, only their combination with T-optimized small-field decompositions and the resulting numbers.

VI. HOW THE RESULTS WERE FOUND

The work used a multi-agent system of five language-model agents (Claude Sonnet 5.5, Anthropic), run as non-interactive coding sessions on one laptop. A supervising session (Claude Opus 5.5) coordinated them. The agents shared a message board and could not modify the verifier, the record table or the board software. Every submission was frozen and verified, and the verifier, not the submitting agent, posted the verdict. The roles were TODD-type reduction, construction from bilinear formulas, local search in the coset of decompositions, large fields, and audit.

The run shows why verification has to cover the record table as well as the arithmetic. Our initial records came from Refs. [5, 6]. Six minutes after launch the auditor agent found Ref. [8], which had lowered several of them. The table was updated and re-frozen, and two wins certified in the minutes between the alert and the update, against the old values ($n = 7$: 103; $n = 10$: 193), were revoked as superseded. The auditor also found three robustness bugs in the new verifier and gaps in the board's identity model. None could produce a false win, and all were fixed before the results reported here.

The subfield idea came from the large-field agent 29 minutes after launch, with Karatsuba over subfields ($n = 16$: $9 \times 39 = 351$). The coordinating session then pointed out the interpolation generalization of Corollary 1, and the same agent implemented it. The six records in Table I were certified within 27 minutes of the first subfield win.

VII. OUTLOOK

Three directions follow directly. First, TODD-type post-reduction of every interpolation circuit already gave six gates at $n = 16$, and it remains to be applied at $n = 8$, 12, 14, 15 and 32 (the last needs a reducer for more than 64 variables). Second, any improvement of the small bases $\mathcal{T}(3)$, $\mathcal{T}(4)$, $\mathcal{T}(5)$ multiplies through Theorem 1; obtaining a certified 23-gate GF(8) decomposition would give $\mathcal{T}(9) \le 115$ and $\mathcal{T}(12) \le 161$. Third, the same composition applies to Toffoli counts and to other bilinear maps, and lower bounds beyond $n = 2$ remain open. Exact enumeration is infeasible already at $n = 3$, where $\dim C = 382$.

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DATA AVAILABILITY

All six record decompositions, the verifier, its tests, the independent checkers and the construction scripts are in the project repository. The interpolation and Karatsuba circuits are reproduced deterministically by `work/large/interp.py` and `tower.py` from the base decompositions in about 0.1 s. The 267-gate $n = 16$ circuit and the 103-gate GF(2^7) base come from seeded FastTODD-type runs, and their outputs are stored.

Appendix A: Construction data

Each circuit is specified by its modulus P_n, its subfield $K = \mathrm{GF}(2^m)$ (generated by a root of the base modulus), the extension degree d, and its base decomposition. Finite interpolation points are the first $2d-2$ elements of K in its coordinate order. The decompositions themselves, lists of $R\mathcal{T}(m)$ vectors in $\mathbb{F}_2^{3n}$, are in the repository.

TABLE III. Moduli and bases of the certified circuits.

n	P_n	(m, d)	base
8	$x^8 + x^4 + x^3 + x + 1$	(4,2)	ATQ GF(16), 39
12	$x^{12} + x^3 + 1$	(4,3)	ATQ GF(16), 39
14	$x^{14} + x^5 + 1$	(7,2)	our GF(2^7), 103
15	$x^{15} + x + 1$	(3,5)	ATQ GF(8), 29
16	$x^{16} + x^5 + x^3 + x + 1$	(4,4)	ATQ GF(16), 39
32	$x^{32} + x^7 + x^3 + x^2 + 1$	(4,8)	ATQ GF(16), 39

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